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Workshop 3: LU Factorization

1. Determine the matrix that implements the transformation $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} x_1 \\ 2x_2 \\ x_3 \end{bmatrix}$.
2. Determine the elementary matrix in $\mathbb{R}^{3\times 3}$ that implements the row operations $R_3 \leftarrow R_3 - 2R_1$.
3. Determine the elementary matrix in $\mathbb{R}^{4\times 4}$ that implements the row operations $R_4 \leftrightarrow R_2$.
4. Determine the matrix in $\mathbb{R}^{3\times 3}$ that implements the following sequence of row operations.

$$\begin{aligned} R_2 &\leftarrow 3R_2 \\ R_1 &\leftarrow R_1 - 2R_2 \\ R_2 &\leftrightarrow R_3 \end{aligned}$$

5. Let E denote the matrix you got from the previous part and let A denote the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$. Determine EA by matrix multiplication. Also apply the row operations from the previous problem to A. Verify that you get the same thing.
6. Let A denote the matrix $\begin{bmatrix} 2 & 1 & 2 & 5 \\ 4 & 2 & 8 & 7 \\ -2 & -1 & 2 & -1 \end{bmatrix}$ and let U denote the matrix $\begin{bmatrix} 2 & 1 & 2 & 5 \\ 0 & 0 & 4 & -3 \\ 0 & 0 & 0 & 7 \end{bmatrix}$ which is an echelon form of A.
   (a) Determine the sequence of row operations used in forward elimination (using only replacements) to transform A to U.
   (b) Determine the matrix that implements the row operations from part (a).
   (c) Determine the sequence of row operations that "inverts" the operations from part (a), i.e., that transforms U to A.
   (d) Determine the inverse of the matrix from part (c).
   (e) Determine an LU-factorization of A.
   (f) Describe the relationship between the row operations in part (a) and the entries of the matrix L from part (d).

Write your solutions below and on the back on this page.

1. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

2. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$

3. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$

4. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \leftarrow 3R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$\xrightarrow{R_1 \leftarrow R_1 - 2R_2} \begin{bmatrix} 1 & -6 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$\xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & -6 & 0 \\ 0 & 0 & 1 \\ 0 & 3 & 0 \end{bmatrix}$

5. $\begin{bmatrix} 1 & -6 & 0 \\ 0 & 0 & 1 \\ 0 & 3 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} -23 & -28 & -33 \\ 7 & 8 & 9 \\ 12 & 15 & 18 \end{bmatrix}$ = ✓

$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \xrightarrow{R_2 \leftarrow 3R_2} \begin{bmatrix} \overset{-24}{1} & \overset{-30}{2} & \overset{-36}{3} \\ 12 & 15 & 18 \\ 7 & 8 & 9 \end{bmatrix} \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \begin{bmatrix} -23 & -28 & -33 \\ 12 & 15 & 18 \\ 7 & 8 & 9 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} -23 & -28 & -33 \\ 7 & 8 & 9 \\ 12 & 15 & 18 \end{bmatrix}$

Continue your work here.

6\. a)

$$\begin{bmatrix} 2 & 1 & 2 & 5 \\ 4 & 2 & 8 & 7 \\ -2 & -1 & 2 & -1 \end{bmatrix} \xrightarrow[R_3 \leftarrow R_3 + R_1]{R_2 \leftarrow R_2 + 2R_1} \begin{bmatrix} 2 & 1 & 2 & 5 \\ 0 & 0 & 4 & -3 \\ 0 & 0 & 4 & 4 \end{bmatrix}$$

$$R_2 \leftarrow R_2 \boxed{+2R_1}$$
$$R_3 \leftarrow R_3 \boxed{+R_1}$$
$$R_3 \leftarrow R_3 \boxed{-R_2}$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{bmatrix} 2 & 1 & 2 & 5 \\ 0 & 0 & 4 & -3 \\ 0 & 0 & 0 & 7 \end{bmatrix}$$

b)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}$$

c)

$$R_3 \leftarrow R_3 + R_2$$
$$R_3 \leftarrow R_3 - R_1$$
$$R_2 \leftarrow R_2 - 2R_1$$

d)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ \boxed{-2} & 1 & 0 \\ \boxed{-1} & \boxed{1} & 1 \end{bmatrix}$$

e)

$$A = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 2 & 1 & 2 & 5 \\ 0 & 0 & 4 & -3 \\ 0 & 0 & 0 & 7 \end{bmatrix}}_{U}$$

f) $R_i \leftarrow R_i + c R_j \implies L_{ij} = -c$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$